OVER A RING WHOSE LOCALIZATIONS SATISFY THE WARFIELD CONDITIONS EVERY FINITELY PRESENTED MODULE IS A SUMMAND OF A FINITE SUM OF QUOTIENTS BY PRINCIPAL IDEALS

THE STACKS PROJECT AUTHORS

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Editorial note. The counted statement span includes the author's own \begin{reference} block citing the work in which the result appears; that block is delivered verbatim and, because the project defines the reference environment as a comment environment, it produces no unresolved citation. The author's typo 'this means there is an exact sequence' region and the phrase 'we see that M_m is a direct sum' are reproduced unchanged.

Allowed standard prerequisites (editorial).

- The characterisation of modules that are summands of direct sums of modules R/fR. invoked at proof lines 37756 and 37777. Converts the required summand statement into the surjectivity of a Hom map for all short exact sequences of the stated form.
- Localization of modules and of Hom out of a finitely presented module. used at proof lines 37758-37772. Lets surjectivity be checked at maximal ideals and identifies the localized Hom modules.
- The structure of finitely presented modules over a ring whose localizations satisfy the generalized valuation ring conditions. invoked at proof lines 37773-37776. Supplies the direct sum decomposition of M_m that makes the localized map surjective.

Source reference. Original lemma 00HN: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 649–649. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 0583: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6766–6766. Cross-chapter navigation pointer into

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the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 37586: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37586–37586. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 37638: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37638–37638. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 37675: sources/stacks-more-algebra/pinned-original-source.tex, source lines 37675–37675. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring such that every local ring of R at a maximal ideal satisfies the equivalent conditions of Lemma 37638. Then every finitely presented R-module is a summand of a finite direct sum of modules of the form R/fR for f in R varying.*

Proof. Let M be a finitely presented R-module. We first show that M is a summand of a direct sum of modules of the form R/fR and at the end we argue the direct sum can be taken to be finite. Let

$$0 \to A \to B \to C \to 0$$

be a short exact sequence of R-modules such that $fA = A \cap fB$ for all $f \in R$. By Lemma 37586 we have to show that $\mathrm{Hom}_R(M, B) \to \mathrm{Hom}_R(M, C)$ is surjective. It suffices to prove this after localization at maximal ideals $\mathfrak{m}$, see Algebra, Lemma 00HN. Note that the localized sequences $0 \to A_\mathfrak{m} \to B_\mathfrak{m} \to C_\mathfrak{m} \to 0$ satisfy the condition that $fA_\mathfrak{m} = A_\mathfrak{m} \cap fB_\mathfrak{m}$ for all $f \in R_\mathfrak{m}$ (because we can write $f = uf'$ with $u \in R_\mathfrak{m}$ a unit and $f' \in R$ and because localization is exact). Since M is finitely presented, we see that

$$\mathrm{Hom}_R(M, B)_\mathfrak{m} = \mathrm{Hom}_{R_\mathfrak{m}}(M_\mathfrak{m}, B_\mathfrak{m}) \quad \text{and} \quad \mathrm{Hom}_R(M, C)_\mathfrak{m} = \mathrm{Hom}_{R_\mathfrak{m}}(M_\mathfrak{m}, C_\mathfrak{m})$$

by Algebra, Lemma 0583. The module $M_\mathfrak{m}$ is a finitely presented $R_\mathfrak{m}$-module. By Lemma 37675 we see that $M_\mathfrak{m}$ is a direct sum of modules of the form $R_\mathfrak{m}/fR_\mathfrak{m}$. Thus we conclude by Lemma 37586 that the map on localizations is surjective.

At this point we know that M is a summand of $\bigoplus_{i \in I} R/f_iR$. Consider the map $M \to \bigoplus_{i \in I} R/f_iR$. Since M is a finite R-module, the image is contained in $\bigoplus_{i \in I'} R/f_iR$ for some finite subset $I' \subset I$. This finishes the proof. □